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1 SAP Connectivity in Microsoft Fabric

Last updated: April 2026
Sources: Microsoft Fabric Documentation, Ignite 2025, FabCon 2026

1.1 Overview

Microsoft Fabric offers multiple ways to connect to SAP systems, ranging from traditional batch/ETL connectors in Data Factory to near real-time replication via Mirroring. The right approach depends on freshness requirements, SAP source system, availability of SAP Datasphere, and the desired analytics pattern.

Power BI Direct Lake (GA March 2026): Data ingested into OneLake – whether via connectors, Copy Job, or Mirroring – can be consumed by Power BI using Direct Lake mode. This eliminates the need for data import or intermediate semantic models, ensuring dashboards reflect the latest data with near-in-memory performance.

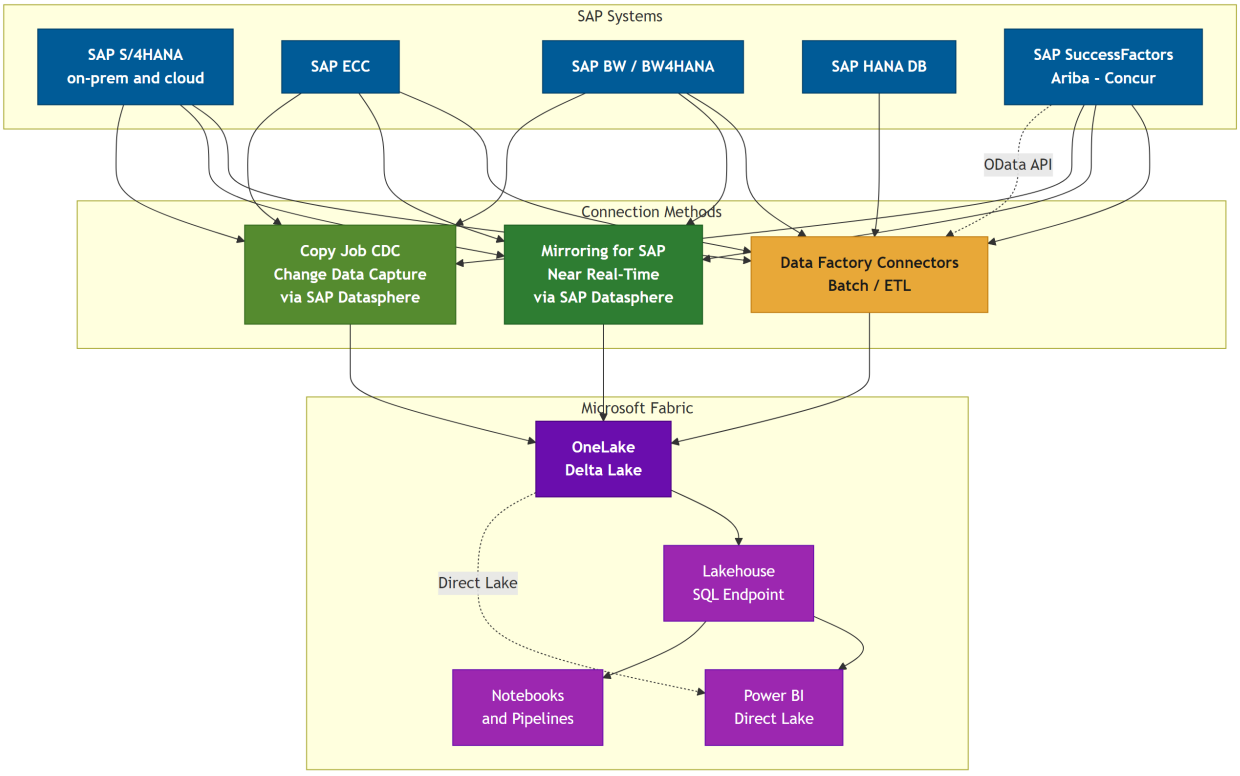

Figure 1: Diagram 1

1.2 Method 1 – Data Factory Connectors (Batch / ETL)

Seven dedicated SAP connectors are available in Microsoft Fabric Data Factory for scheduled or on-demand data extraction. An **OData connector** is also available for SAP SaaS applications (Suc-

cessFactors, S/4HANA Cloud, C4C) that expose OData APIs.

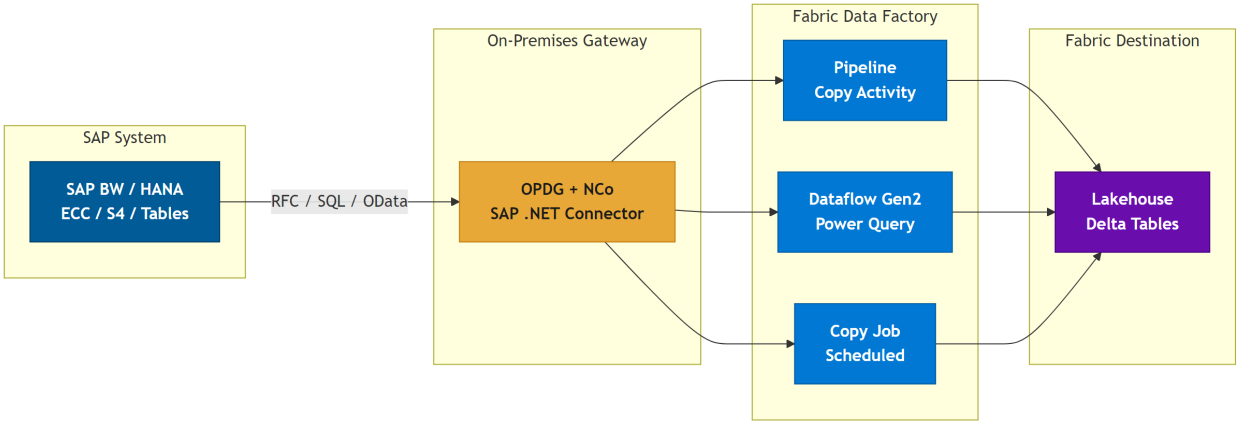

Figure 2: Diagram 2

1.2.1 Connector Reference Table

Connector	Dataflow Gen2	Pipeline (Copy)	Copy Job	Gateway
SAP BW Application Server	● Import + DQ	○	○	OPDG + NCo 3.x
SAP BW Message Server	● Import + DQ	○	○	OPDG + NCo 3.x
SAP BW Open Hub – App Server	●	●	○	OPDG
SAP BW Open Hub – Msg Server	●	●	○	OPDG
SAP HANA Database	● incl. DQ	● Lookup + Copy	●	OPDG
SAP Table – App Server	○	●	●	OPDG + NCo
SAP Table – Msg Server	○	●	○	OPDG
OData (generic)	●	●	○	None / OPDG

Legend: ● = Supported | ○ = Not supported | DQ = DirectQuery

1.2.2 When to Use

- **SAP BW connectors** – BW InfoProviders, BEx queries, Open Hub destinations. BW 7.3, 7.5, BW/4HANA 2.0. Best for aggregated datasets.
- **SAP HANA** – HANA views, tables, stored procedures. Also supports DirectQuery via Dataflow Gen2.

- **SAP Table** – ABAP table/view extraction via RFC (`VBAK`, `MARA`, `KNA1`). Medium-volume extractions.
- **OData** – SAP apps exposing OData APIs (SuccessFactors, S/4HANA Cloud, C4C). Small-to-medium volumes.

1.2.3 Infrastructure Prerequisites

All SAP connectors (except OData) require this infrastructure, **even when SAP runs in the cloud**:

1. **On-Premises Data Gateway (OPDG)** deployed near the SAP system
2. **SAP .NET Connector (NCo) 3.0 or 3.1** installed on the gateway server
3. **Network access** to SAP: RFC ports (33XX), HANA port (30015), via VPN/VNet or ExpressRoute
4. **SAP technical account** with appropriate authorizations (`S_RFC`, `S_TABU_DIS`)

Important limitations:

- **No native CDC** – full or watermark-based incremental only. SAP deletions are not tracked.
- **No dedicated connector for SAP SaaS** – SuccessFactors, Ariba, Concur require OData or Mirroring/CDC.
- **Performance impact on SAP** – large RFC/BW extractions consume SAP resources. For massive tables, prefer HANA direct or Mirroring.

1.3 Method 2 – Mirroring for SAP (Near Real-Time)

Mirroring for SAP provides **continuous, near real-time replication** of SAP data into OneLake, without any custom ETL. It operates as a Fabric item ("mirrored database"), fully managed by the platform.

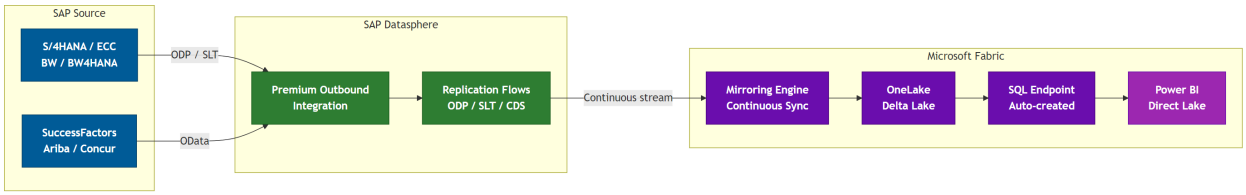

Figure 3: Diagram 3

1.3.1 Supported SAP Sources

SAP System	Deployment	Supported
SAP S/4HANA	On-premises	●
SAP S/4HANA Cloud	Cloud (public + private)	●
SAP ECC	On-premises	●

SAP System	Deployment	Supported
SAP BW	On-premises	●
SAP BW/4HANA	On-premises and cloud	●
SAP SuccessFactors	SaaS	●
SAP Ariba	SaaS	●
SAP Concur	SaaS	●

Other SAP systems (CRM, SRM, SCM) based on NetWeaver ABAP are covered via ODP/SLT, same as ECC. Any SAP system supporting ODP extraction is eligible.

1.3.2 Key Benefits

- **Zero ETL** – schema evolution handled automatically; raw replication (transforms applied downstream)
- **Near real-time** – latency typically seconds to a few minutes
- **End-to-end lineage** – full governance and audit trail
- **Native Fabric integration** – SQL endpoint, Power BI Direct Lake, Notebooks, Lakehouses
- **Minimal impact on SAP** – uses standard ODP/SLT mechanisms via Datasphere (not custom queries)
- **Up to 1,000 tables** per mirrored database (increased from ~100 at FabCon 2026)

1.3.3 Prerequisites

1. **SAP Datasphere** with **Premium Outbound Integration** (mandatory)
2. Replication Flows configured in Datasphere
3. **On-premises SAP:** Data Provisioning Agent or SAP Cloud Connector on-site
4. **SAP Cloud sources:** OData connections activated and registered in Datasphere
5. Fabric capacity (F2+ recommended)
6. Network: SAP Datasphere to Fabric (outbound HTTPS)

SAP licensing note: SAP requires official extraction products (Datasphere, Data Intelligence) for ODP-based extraction. This reinforces Mirroring as the strategic SAP-endorsed path.

1.4 Method 3 – Copy Job CDC for SAP

Introduced at **Ignite 2025**, Copy Job supports **Change Data Capture (CDC)** for SAP via Datasphere. Unlike Mirroring (autonomous), Copy Job CDC provides **explicit orchestration control** within a Data Factory pipeline.

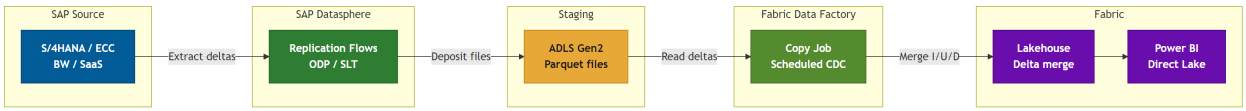

Figure 4: Diagram 4

1.4.1 How It Works

Two-stage mechanism:

1. **SAP Datasphere** extracts initial data then delta changes (ODP/SLT) and deposits Parquet files on **Azure Data Lake Storage Gen2**.
2. **Fabric Copy Job** reads those files and merges inserts/updates/deletes into the Fabric Lakehouse (Delta).

1.4.2 Feature Summary

Feature	Value
Change types captured	Inserts, Updates, Deletes
Watermark column needed	○ Not required
Manual refresh needed	○ Scheduled trigger
Merge destination	Lakehouse (Delta)
Intermediate storage	ADLS Gen2 / S3 / GCS

1.4.3 Prerequisites

1. **SAP Datasphere** with **Premium Outbound Integration** (same as Mirroring)
2. Data Provisioning Agent for on-premises sources
3. Replication Flows targeting a cloud storage container (ADLS Gen2)
4. Fabric Copy Job configured to read from that container

1.4.4 When to Prefer over Mirroring

- **Control the schedule** (e.g., every 15 min during business hours, pause at night)
- **Multi-source pipeline** with additional transforms, validations, or joins
- **Limit continuous load** – scheduled bursts vs. 24/7 streaming
- **Monitoring** is split: Datasphere (replication health) + Fabric (Copy Job runs)

> **Latency** depends on scheduled frequency. A 5-minute interval = up to 5 minutes stale. For continuous freshness, use Mirroring.

1.4.5 Copy Job Optimizations (FabCon 2026)

- **Auto-partitioning** for better performance on large copies

- **Automatic audit columns** for load tracking
- **Zero CU cost** when no data changes exist

1.5 Alternative Approaches

1.5.1 OneLake Shortcuts

If SAP data already exists in external storage (ADLS, S3), create a **OneLake Shortcut** to make it available in Fabric without re-copying.

1.5.2 Third-Party ETL Tools

Informatica, Boomi, Theobald, and others offer SAP connectors writing to OneLake. Microsoft's strategic direction favors native connectors and SAP Datasphere.

1.5.3 Logic Apps / Power Automate

For event-driven micro-integrations (e.g., SAP order creation triggers a Fabric action). Not suitable for bulk data.

1.6 Decision Guide

1.7 Key Announcements

1.7.1 Ignite 2025 – November 2025

Feature	Status	Coverage
Mirroring for SAP	◻ Preview	S/4HANA, BW, BW/4HANA, SuccessFactors, Ariba
Copy Job CDC for SAP	● GA	SAP via Datasphere to Lakehouse

1.7.2 FabCon 2026 – March 2026

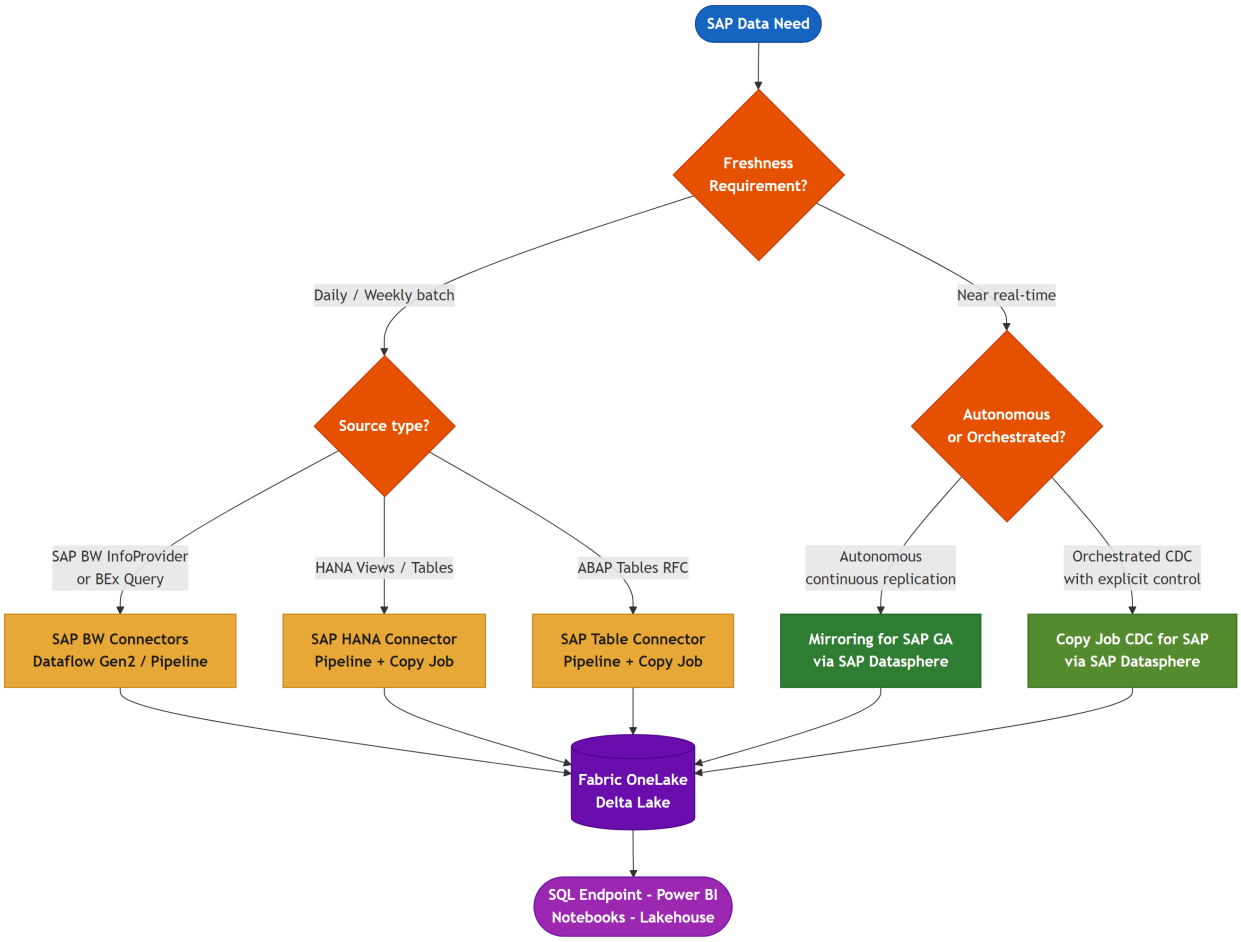

Figure 5: Diagram 5

Feature	Status	What's New
Mirroring for SAP	● GA	+ SAP ECC, + Concur. Up to 1,000 tables.
Copy Job enhancements	● GA	Auto-partitioning, audit columns, zero-cost
Direct Lake for Power BI	● GA	Dashboards read Delta directly from OneLake

Docs: Microsoft Fabric Mirrored Databases From SAP

1.8 Comparison Summary

Criteria	Batch Connectors	Copy Job CDC	Mirroring for SAP
Freshness	Hourly to daily	Minutes (scheduled)	Near real-time (continuous)
Custom ETL needed	Required	Minimal	None (zero-ETL)
SAP Datasphere	○ Not required	● Required	● Required
Intermediate storage	○	● ADLS Gen2	○ Direct to OneLake
SAP sources	BW, HANA, Tables	Full landscape	Full landscape
Power BI access	DQ (BW/HANA) + Import	Direct Lake	SQL Endpoint + Direct Lake
Native CDC	○ Not supported	● Scheduled	● Continuous
Max tables	Per pipeline	Per job	1,000 per mirrored DB
In-flight transform	Via Dataflow Gen2	Post-copy	○ Raw replication
GA status	All GA (2023)	GA (Nov 2025)	GA (March 2026)

Legend: ● = Yes / Supported | ○ = No / Not supported | ◻ = Preview

1.9 Recommendations by Scenario

Historical bulk load (migrating years of data): Use **batch connectors** – SAP HANA or SAP Table via Pipeline Copy job with partitioning.

Regular analytics refresh (daily/hourly dashboards): Use **batch connectors** for simple cases, or **Copy Job CDC** for incremental deltas.

Real-time operational analytics (live sales, inventory): Use **Mirroring for SAP** + Power BI Direct Lake for continuous dashboard freshness.

SAP SaaS without Datasphere (SuccessFactors, Ariba): Use the **OData connector** for moderate volumes; invest in Datasphere for scale.

Multi-source orchestrated pipeline (SAP + other sources): Use **Copy Job CDC** in a Data Factory pipeline with transforms and validations.

No SAP Datasphere available: Use **batch connectors** + OPDG. Consider **OneLake Shortcuts** for existing data lakes.

1.10 Appendix A – References

1.10.1 Mirroring for SAP

Resource	Link
Mirrored Databases from SAP	https://learn.microsoft.com/fabric/mirroring/sap
Mirroring Overview	https://learn.microsoft.com/fabric/mirroring/overview
Extended Capabilities (CDF, Views)	https://learn.microsoft.com/fabric/mirroring/extended-capabilities
Troubleshooting Guide	https://learn.microsoft.com/fabric/mirroring/troubleshooting

1.10.2 Data Factory SAP Connectors

Resource	Link
Connector Overview (all)	https://learn.microsoft.com/fabric/data-factory/connector-overview
SAP BW Open Hub	https://learn.microsoft.com/fabric/data-factory/connector-sap-bw-open-hub-overview

Resource	Link
SAP HANA	https://learn.microsoft.com/fabric/data-factory/connector-sap-hana-database-overview
SAP Table	https://learn.microsoft.com/fabric/data-factory/connector-sap-table-overview
SAP BW Application Server	https://learn.microsoft.com/power-query/connectors/sap-bw/application-setup-and-connect
OData Connector	https://learn.microsoft.com/fabric/data-factory/connector-odata-overview

1.10.3 Copy Job and CDC

Resource	Link
What is Copy Job	https://learn.microsoft.com/fabric/data-factory/what-is-copy-job
CDC in Copy Job	https://learn.microsoft.com/fabric/data-factory/copy-job-change-data-capture
Copy Job Monitoring	https://learn.microsoft.com/fabric/data-factory/copy-job-workspace-monitoring

1.10.4 Power BI and OneLake

Resource	Link
Direct Lake Mode	https://learn.microsoft.com/fabric/fundamentals/direct-lake-overview
OneLake Shortcuts	https://learn.microsoft.com/fabric/onelake/onelake-shortcuts

1.10.5 Announcements

Resource	Link
Ignite 2025 Feature Summary	https://blog.fabric.microsoft.com/en-us/blog/fabric-november-2025-feature-summary
FabCon 2026 Feature Summary	https://blog.fabric.microsoft.com/en-us/blog/fabric-march-2026-feature-summary
FabCon 2026 Hero Blog	https://aka.ms/FabCon-SQLCon-2026-news
SAP in Fabric Blog (Sept 2024)	https://blog.fabric.microsoft.com/en-us/blog/connecting-to-sap-data-in-microsoft-fabric

1.10.6 SAP Datasphere

Resource	Link
SAP Datasphere Docs	https://help.sap.com/docs/SAP_DATASPHERE
Premium Outbound Integration	https://help.sap.com/docs/SAP_DATASPHERE/be5967d099974c69b77f4549425ca4c0/eb7ff31
Data Provisioning Agent	https://help.sap.com/docs/SAP_DATASPHERE/935116dd7c324355803d4b85809cec97

1.10.7 Infrastructure

Resource	Link
On-Premises Data Gateway	https://learn.microsoft.com/data-integration/gateway/service-gateway-onprem
SAP .NET Connector (NCo)	https://support.sap.com/en/product/connectors/msnet.html

1.11 Appendix B – Glossary

Acronym	Definition
CDC	Change Data Capture – track inserts, updates, and deletes
CDS	Core Data Services – SAP data modeling framework
ODP	Operational Data Provisioning – SAP delta extraction framework
SLT	SAP Landscape Transformation – trigger-based real-time replication
OPDG	On-Premises Data Gateway – Microsoft gateway for on-prem data
NCo	SAP .NET Connector – library for SAP RFC communication
RFC	Remote Function Call – SAP native inter-system protocol
Direct Lake	Power BI mode reading Delta files directly from OneLake
DQ	DirectQuery – live query mode in Power BI